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0 Introduction

Ce rapport présente les résultats d’une analyse approfondie de plusieurs stratégies de construction de portefeuilles sur des données d’actions américaines issues de la bibliothèque Kenneth French. Quatre jeux de données sont utilisés : 25 portefeuilles formés sur la taille et le ratio book-to-market (5×5), 100 portefeuilles (10×10), 10 portefeuilles sectoriels et 48 portefeuilles sectoriels. Pour chaque jeu de données, les fréquences **mensuelle** et **quotidienne** sont exploitées sur la période **janvier 1970 – décembre 2025**.

Les rendements sont exprimés en proportion (divisés par 100) et les données manquantes sont remplacées par zéro. Les fenêtres glissantes (rolling) utilisées en *out-of-sample* comprennent une fenêtre d’estimation de **10 ans** (120 mois ou 2520 jours) et un pas de **6 mois** (6 mois ou 126 jours).

L’optimisation de portefeuille souffre de deux pathologies majeures dans les situations de haute dimensionnalité :

- **L’erreur d’estimation** : les paramètres μ et Σ sont estimés sur des données historiques, ce qui introduit un biais qui se traduit par une dégradation des performances *out-of-sample*.
- **La malédiction de la dimensionnalité** : lorsque N est grand relativement à T , la matrice de covariance estimée devient mal conditionnée, rendant son inversion instable.

Les sections suivantes décrivent comment les différentes stratégies abordent ces défis.

1 Baseline Strategies : MSR, MinVar and Equal Weight

1.1 Methodology

We compare three standard portfolio strategies :

1. **Maximum Sharpe Ratio portfolio (MSR)** :

$$w_{\text{MSR}} = \frac{\Sigma^{-1}\mu}{\mathbf{1}^\top \Sigma^{-1}\mu}.$$

2. **Minimum-variance portfolio (MinVar)** :

$$w_{\text{MinVar}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

3. **Equally weighted portfolio (EW)** :

$$w_{\text{EW}} = \frac{1}{N}\mathbf{1}.$$

The *in-sample* portfolios are computed using the full sample. The *out-of-sample* analysis

follows a rolling-window procedure : a 10-year estimation window is used to compute the portfolio weights, which are then applied to the following 6 months. The window is then rolled forward by 6 months until the end of the sample.

Performance is reported using annualized mean return, annualized volatility and annualized Sharpe ratio :

$$\bar{r}_{\text{ann}} = \bar{r} \times f, \quad \sigma_{\text{ann}} = \hat{\sigma} \times \sqrt{f}, \quad \text{SR} = \frac{\bar{r}_{\text{ann}} - r_f}{\sigma_{\text{ann}}},$$

where $f = 12$ for monthly data and $f = 252$ for daily data. Throughout the analysis, we set $r_f = 0$, so that $\text{SR} = \bar{r}_{\text{ann}}/\sigma_{\text{ann}}$. This allows us to compare the strategies directly in terms of risk-adjusted raw returns.

1.2 In-Sample Results

TABLE 1 – *In-sample* performance – monthly and daily data

Frequency	Dataset	Strategy	Ann. Mean	Ann. Vol.	Sharpe
Monthly	25 Ptf (5×5)	MSR	28.57%	17.02%	1.67
		MinVar	14.97%	12.32%	1.21
		EW	13.15%	18.22%	0.72
	100 Ptf (10×10)	MSR	47.70%	19.38%	2.46
		MinVar	15.16%	10.92%	1.38
		EW	13.22%	18.42%	0.72
	10 Industries	MSR	12.36%	12.72%	0.97
		MinVar	11.61%	12.33%	0.94
		EW	12.10%	15.04%	0.81
	48 Industries	MSR	23.54%	16.42%	1.43
		MinVar	10.62%	11.03%	0.96
		EW	12.18%	17.06%	0.71
Daily	25 Ptf (5×5)	MSR	32.88%	18.53%	1.77
		MinVar	16.01%	12.93%	1.24
		EW	12.91%	17.48%	0.74
	100 Ptf (10×10)	MSR	41.82%	16.54%	2.53
		MinVar	17.48%	10.69%	1.64
		EW	12.99%	17.50%	0.74
	10 Industries	MSR	12.30%	13.62%	0.90
		MinVar	11.56%	13.20%	0.88
		EW	12.17%	16.01%	0.76
	48 Industries	MSR	23.31%	17.96%	1.30
		MinVar	10.57%	12.09%	0.87
		EW	12.08%	16.69%	0.72

1.3 Out-of-Sample Results

TABLE 2 – Out-of-sample performance – monthly and daily data

Frequency	Dataset	Strategy	Ann. Mean	Ann. Vol.	Sharpe	Turnover
Monthly	25 Ptf	MSR	26.01%	25.23%	1.03	5.93
		MinVar	17.91%	13.55%	1.32	2.02
		EW	13.55%	17.62%	0.77	0.00
	100 Ptf	MSR	150.31%	1551.41%	0.10	538.39
		MinVar	18.14%	25.57%	0.71	16.72
		EW	13.63%	17.80%	0.76	0.00
	10 Industries	MSR	11.99%	17.15%	0.70	1.23
		MinVar	12.17%	12.40%	0.98	0.39
		EW	13.12%	14.70%	0.89	0.00
	48 Industries	MSR	14.49%	42.32%	0.34	8.14
		MinVar	9.97%	14.23%	0.70	1.97
		EW	13.03%	16.40%	0.79	0.00
Daily	25 Ptf	MSR	27.71%	20.69%	1.34	2.95
		MinVar	19.87%	13.16%	1.51	0.66
		EW	13.59%	18.30%	0.74	0.00
	100 Ptf	MSR	33.63%	23.40%	1.44	5.01
		MinVar	19.51%	11.06%	1.76	0.98
		EW	13.67%	18.31%	0.75	0.00
	10 Industries	MSR	11.24%	21.55%	0.52	1.42
		MinVar	12.24%	13.40%	0.91	0.20
		EW	13.33%	16.62%	0.80	0.00
	48 Industries	MSR	12.06%	36.00%	0.34	4.28
		MinVar	11.11%	12.41%	0.89	0.46
		EW	13.19%	17.22%	0.77	0.00

1.4 Discussion

The *in-sample* results strongly favour the MSR portfolio, especially for the 100 size-book-to-market portfolios, where the Sharpe ratio reaches 2.46 with monthly data and 2.53 with daily data. However, this performance is mostly driven by the use of sample estimates of expected returns, which are very noisy and lead to overfitting.

The *out-of-sample* results are more informative. The MSR strategy becomes unstable in several datasets, particularly in high dimension. The most extreme case is the 100 monthly portfolios, where volatility reaches 1551.41%, the Sharpe ratio drops to 0.10, and turnover increases to 538.39. This confirms that small estimation errors in expected returns can generate very large changes in portfolio weights.

By contrast, MinVar is more robust because it does not rely on expected-return estimates. It often achieves better out-of-sample Sharpe ratios than MSR, with lower volatility and turnover. For example, with daily data on the 100 portfolios, MinVar reaches a Sharpe ratio of 1.76, compared with 1.44 for MSR and 0.75 for EW.

The EW portfolio remains a strong benchmark. It is simple, stable across datasets and frequencies, and has zero turnover in our implementation. This makes it attractive when transaction costs are considered.

Overall, the results show that dimensionality and estimation error are the main drivers of instability. Moving from 25 to 100 portfolios, or from 10 to 48 industries, makes optimized strategies less reliable, especially MSR. Daily data generally improve covariance estimation and reduce instability, but they do not fully solve the problem of estimating expected returns. Therefore, simple or variance-based strategies are more reliable out-of-sample than aggressive mean-variance optimization.

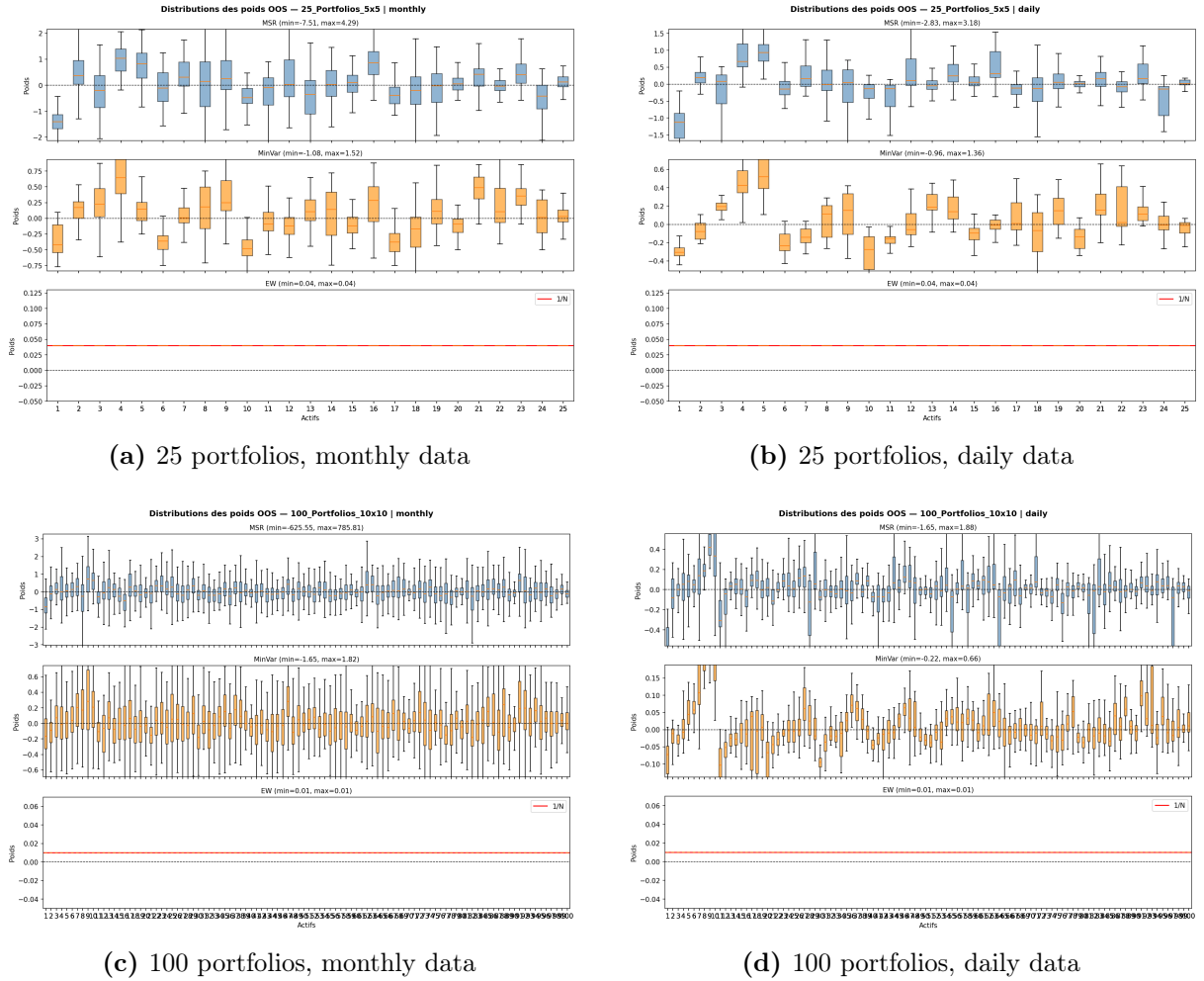


FIGURE 1 – Boxplots of *out-of-sample* portfolio weights for the 25- and 100-portfolio datasets, using monthly and daily data.

2 MinVar Portfolio with ℓ_2 -Norm Regularization

2.1 Methodology

We now consider the ℓ_2 -regularized minimum-variance portfolio, defined as

$$\mathbf{w}_\gamma = \operatorname{argmin}_{\mathbf{1}^\top \mathbf{w} = 1} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} + \gamma \mathbf{w}^\top \mathbf{w}.$$

This problem has the following closed-form solution :

$$\mathbf{w}_\gamma = \frac{\tilde{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \tilde{\Sigma}^{-1}\mathbf{1}}, \quad \tilde{\Sigma} = \hat{\Sigma} + \gamma I_N.$$

In the implementation, γ is normalized as

$$\gamma_{\text{eff}} = \gamma_{\text{raw}} \times \frac{\text{tr}(\hat{\Sigma})}{N},$$

so that the penalty remains comparable across datasets with different dimensions and volatility levels. The tested grid is

$$\gamma \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 0.7, 1.0\}.$$

The role of the ℓ_2 penalty is threefold. First, adding γI_N improves the conditioning of the covariance matrix and makes the inversion more stable. Second, it shrinks the portfolio weights toward the equally weighted portfolio. Third, it penalizes concentrated allocations, thereby reducing extreme long and short positions. This should improve robustness and reduce turnover, especially when the number of assets is large relative to the estimation window.

2.2 Effect of γ on Out-of-Sample Performance

The following tables report the out-of-sample performance for each dataset and frequency. The columns show the annualized mean return, annualized volatility, Sharpe ratio with $r_f = 0$, and average turnover.

TABLE 3 – Sensitivity to γ – 25 portfolios (5×5), monthly and daily data

	Monthly ($T = 672$, $N = 25$)				Daily ($T = 14121$, $N = 25$)			
γ	Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
MinVar (0)	17,91%	13,55%	1,322	2,016	19,87%	13,16%	1,510	0,661
0,001	17,85%	13,45%	1,328	1,926	19,82%	13,13%	1,510	0,647
0,002	17,79%	13,36%	1,332	1,847	19,77%	13,10%	1,510	0,634
0,005	17,64%	13,17%	1,340	1,658	19,64%	13,02%	1,508	0,600
0,01	17,44%	12,97%	1,345	1,438	19,44%	12,94%	1,502	0,556
0,02	17,15%	12,79%	1,341	1,169	19,13%	12,87%	1,487	0,492
0,05	16,65%	12,73%	1,308	0,801	18,53%	12,92%	1,434	0,383
0,1	16,21%	12,91%	1,255	0,565	17,95%	13,20%	1,360	0,298
0,2	15,74%	13,32%	1,181	0,381	17,27%	13,73%	1,258	0,224
0,5	15,11%	14,10%	1,071	0,215	16,27%	14,79%	1,100	0,145
0,7	14,89%	14,43%	1,032	0,172	15,88%	15,23%	1,043	0,122
1,0	14,66%	14,79%	0,992	0,135	15,48%	15,70%	0,986	0,100

TABLE 4 – Sensitivity to γ – 100 portfolios (10×10), monthly and daily data

γ	Monthly ($T = 672, N = 100$)				Daily ($T = 14121, N = 100$)			
	Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
MinVar (0)	18,14%	25,57%	0,709	16,724	19,51%	11,06%	1,765	0,980
0,001	18,73%	21,49%	0,872	11,464	19,48%	11,05%	1,762	0,972
0,002	19,21%	19,87%	0,967	9,436	19,45%	11,05%	1,760	0,965
0,005	19,88%	17,67%	1,125	6,862	19,38%	11,05%	1,754	0,943
0,01	20,13%	16,14%	1,247	5,191	19,26%	11,04%	1,744	0,910
0,02	19,96%	14,86%	1,344	3,834	19,06%	11,04%	1,726	0,856
0,05	19,13%	13,62%	1,404	2,497	18,62%	11,10%	1,678	0,743
0,1	18,24%	13,06%	1,397	1,761	18,15%	11,23%	1,616	0,629
0,2	17,32%	12,80%	1,353	1,214	17,59%	11,51%	1,528	0,505
0,5	16,23%	12,95%	1,253	0,714	16,78%	12,19%	1,376	0,349
0,7	15,88%	13,13%	1,210	0,581	16,47%	12,54%	1,313	0,299
1,0	15,56%	13,39%	1,162	0,464	16,15%	12,98%	1,244	0,252

TABLE 5 – Sensitivity to γ – 10 industries, monthly and daily data

γ	Monthly ($T = 672, N = 10$)				Daily ($T = 14121, N = 10$)			
	Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
MinVar (0)	12,17%	12,40%	0,982	0,394	12,24%	13,40%	0,913	0,201
0,001	12,17%	12,39%	0,982	0,389	12,24%	13,40%	0,913	0,200
0,002	12,17%	12,38%	0,983	0,386	12,24%	13,39%	0,914	0,198
0,005	12,17%	12,35%	0,985	0,375	12,25%	13,38%	0,915	0,193
0,01	12,17%	12,31%	0,989	0,358	12,26%	13,37%	0,917	0,186
0,02	12,17%	12,25%	0,994	0,331	12,29%	13,35%	0,920	0,175
0,05	12,17%	12,13%	1,004	0,277	12,37%	13,35%	0,927	0,150
0,1	12,19%	12,05%	1,012	0,224	12,49%	13,40%	0,932	0,126
0,2	12,23%	12,03%	1,016	0,169	12,66%	13,58%	0,932	0,099
0,5	12,34%	12,22%	1,010	0,106	12,89%	14,07%	0,916	0,065
0,7	12,40%	12,37%	1,003	0,086	12,96%	14,33%	0,905	0,054
1,0	12,48%	12,57%	0,993	0,069	13,04%	14,63%	0,891	0,044

TABLE 6 – Sensitivity to γ – 48 industries, monthly and daily data

γ	Monthly ($T = 672$, $N = 48$)				Daily ($T = 14121$, $N = 48$)			
	Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
MinVar (0)	9,97%	14,23%	0,701	1,969	11,11%	12,41%	0,895	0,463
0,001	9,99%	14,13%	0,707	1,915	11,12%	12,40%	0,897	0,460
0,002	10,01%	14,03%	0,713	1,865	11,13%	12,40%	0,898	0,457
0,005	10,07%	13,78%	0,731	1,740	11,16%	12,38%	0,901	0,448
0,01	10,16%	13,47%	0,754	1,582	11,20%	12,37%	0,906	0,435
0,02	10,34%	13,07%	0,791	1,369	11,28%	12,34%	0,914	0,413
0,05	10,74%	12,48%	0,861	1,036	11,46%	12,31%	0,931	0,364
0,1	11,17%	12,10%	0,923	0,786	11,66%	12,31%	0,947	0,312
0,2	11,62%	11,86%	0,980	0,567	11,92%	12,40%	0,961	0,253
0,5	12,13%	11,89%	1,020	0,346	12,30%	12,71%	0,968	0,177
0,7	12,26%	12,01%	1,021	0,286	12,43%	12,90%	0,964	0,152
1,0	12,38%	12,22%	1,013	0,231	12,56%	13,16%	0,955	0,128

2.3 Discussion

The effect of the ℓ_2 penalty depends strongly on the dimension of the dataset and on the return frequency. For the 25 portfolios, moderate regularization improves the monthly Sharpe ratio from 1.322 for the unregularized MinVar portfolio to 1.345 at $\gamma = 0.01$, while reducing turnover from 2.016 to 1.438. This confirms the expected stabilizing effect of Ridge regularization.

The improvement is much stronger for the 100 monthly portfolios. Without regularization, the MinVar portfolio is unstable, with a volatility of 25.57%, a Sharpe ratio of 0.709 and a turnover of 16.724. With $\gamma = 0.05$, the Sharpe ratio increases to 1.404, volatility falls to 13.62%, and turnover drops to 2.497. This is the clearest evidence that ℓ_2 regularization is useful when the covariance matrix is noisy and poorly conditioned.

For the 10 industry portfolios, the gains are more limited. Since the number of assets is small, the covariance matrix is already easier to estimate, and regularization mainly produces a modest reduction in turnover. The monthly Sharpe ratio increases only slightly, from 0.982 to a maximum of 1.016 around $\gamma = 0.2$.

For the 48 industries, regularization is again beneficial. In monthly data, the Sharpe ratio rises from 0.701 for MinVar to about 1.02 for γ between 0.5 and 0.7, while turnover declines substantially. This suggests that the unregularized MinVar portfolio is too sensitive to estimation noise and that stronger shrinkage toward EW improves out-of-sample robustness.

Daily data generally require less regularization. The larger number of observations makes the covariance estimate more stable, especially for the 100 portfolios, where the unregularized MinVar strategy already has a high Sharpe ratio of 1.765. In this case, increasing γ mainly

reduces turnover but gradually lowers the Sharpe ratio. Overall, the results confirm the bias–variance trade-off : small to moderate values of γ reduce estimation error and improve stability, while excessive regularization moves the portfolio too close to EW and may reduce performance.

3 Optimal Selection of γ by Cross-Validation

3.1 Selection Method

To select the regularization parameter in a data-driven way, we use a 10-fold cross-validation procedure inside each rolling estimation window. For each candidate value of γ , the portfolio is estimated on the training folds and evaluated on the validation fold. Since the strategy is a regularized minimum-variance portfolio, the criterion minimized in cross-validation is the variance of the validation portfolio returns :

$$\gamma^* = \underset{\gamma \in \Gamma}{\operatorname{argmin}} \frac{1}{10} \sum_{k=1}^{10} \operatorname{Var} \left(\mathbf{w}_{\gamma}^{\top} \mathbf{r}^{(\text{val},k)} \right).$$

The tested grid is

$$\Gamma = \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 0.7, 1.0\}.$$

This procedure is repeated at each rolling-window date, which gives a dynamic sequence of selected values γ_t^* .

3.2 Results

TABLE 7 – MinVar vs ℓ_2 portfolio with dynamic γ^* selection

Dataset	Strategy	Monthly				Daily			
		Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
25 Ptf	MinVar	17,91%	13,55%	1,322	2,016	19,87%	13,16%	1,510	0,661
	$\ell_2 (\gamma^*)$	16,68%	12,71%	1,313	1,170	19,65%	13,01%	1,511	0,642
100 Ptf	MinVar	18,14%	25,57%	0,709	16,724	19,51%	11,06%	1,765	0,980
	$\ell_2 (\gamma^*)$	17,38%	13,05%	1,332	1,671	18,95%	11,02%	1,720	0,950
10 Industries	MinVar	12,17%	12,40%	0,982	0,394	12,24%	13,40%	0,913	0,201
	$\ell_2 (\gamma^*)$	12,27%	12,22%	1,005	0,332	12,43%	13,35%	0,931	0,215
48 Industries	MinVar	9,97%	14,23%	0,701	1,969	11,11%	12,41%	0,895	0,463
	$\ell_2 (\gamma^*)$	11,81%	11,78%	1,003	0,685	11,55%	12,31%	0,939	0,452

Mean = annualized mean return ; Vol. = annualized volatility ; SR = Sharpe ratio with $r_f = 0$; TO = average turnover per rebalancing.

TABLE 8 – Time distribution of selected dynamic γ^* values

Dataset	Frequency	$\gamma_{\min}^*$	$\gamma_{\max}^*$	$\bar{\gamma}^*$
25 Ptf	Monthly	0,010	0,300	0,061
25 Ptf	Daily	0,000	0,100	0,016
100 Ptf	Monthly	0,050	0,700	0,264
100 Ptf	Daily	0,000	0,100	0,036
10 Industries	Monthly	0,010	0,500	0,158
10 Industries	Daily	0,000	0,100	0,031
48 Industries	Monthly	0,100	1,000	0,358
48 Industries	Daily	0,000	0,300	0,046

3.3 Discussion

Cross-validation improves the MinVar portfolio mainly when the covariance matrix is difficult to estimate. The strongest improvement appears for the 100 monthly portfolios : the Sharpe ratio increases from 0.709 to 1.332, while turnover falls from 16.724 to 1.671. The same pattern is observed for the 48 monthly industries, where the Sharpe ratio rises from 0.701 to 1.003 and turnover is divided by almost three. In both cases, the dynamic ℓ_2 penalty strongly stabilizes the portfolio weights.

The gains are smaller for the 25 portfolios and 10 industries. These datasets are lower-dimensional, so the unregularized covariance matrix is less unstable. For example, for the 25 monthly portfolios, cross-validation reduces volatility and turnover, but the Sharpe ratio remains close to the original MinVar value. This suggests that the benefit of regularization is mostly a robustness gain rather than a systematic increase in returns.

The selected values of γ^* confirm this interpretation. Monthly datasets require stronger regularization than daily datasets. For instance, the average selected value is $\bar{\gamma}^* = 0.264$ for the 100 monthly portfolios, compared with only 0.036 for the daily version. With daily data, the covariance matrix is estimated from many more observations, so less shrinkage is needed. Overall, the cross-validation procedure adapts the amount of regularization to the level of estimation noise : high-dimensional monthly datasets receive stronger shrinkage, whereas daily datasets remain closer to the standard MinVar portfolio.

4 Alternative Constraints : ℓ_1 and ℓ_1 - ℓ_∞

4.1 ℓ_1 Penalty

The first alternative is an ℓ_1 -regularized minimum-variance portfolio :

$$\mathbf{w}_\lambda = \operatorname{argmin}_{\mathbf{1}^\top \mathbf{w} = 1} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} + \lambda \|\mathbf{w}\|_1.$$

Compared with the ℓ_2 penalty, the ℓ_1 penalty has a different effect on the portfolio weights.

It controls the total absolute exposure and therefore limits leverage and short positions. It can also produce sparse portfolios, with some weights set close to zero or exactly equal to zero. Since the problem has no simple closed-form solution, it is solved using convex quadratic programming with `cvxpy`. The parameter λ^* is selected by 5-fold cross-validation, minimizing the validation portfolio variance over the grid

$$\lambda \in \{10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}, 10^{-1}\}.$$

4.2 ℓ_1 - ℓ_∞ Constraint

The second alternative is inspired by Xing et al. and combines an ℓ_1 constraint with an ℓ_∞ constraint :

$$\mathbf{w}_{\alpha,c} = \underset{\mathbf{1}^\top \mathbf{w} = 1}{\operatorname{argmin}} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \quad \text{s.t.} \quad \|\mathbf{w}\|_1 + \alpha \|\mathbf{w}\|_\infty \leq c.$$

The ℓ_1 part controls total leverage, while the ℓ_∞ part directly limits the largest absolute position in the portfolio. This combination is designed to reduce both excessive short-selling and excessive concentration in a few assets. The parameters (α^*, c^*) are selected by 5-fold cross-validation over the grid

$$\{(0.5, 1.5), (1.0, 1.5), (1.0, 2.0), (2.0, 3.0)\}.$$

4.3 Comparative Results

The tables below compare four strategies for each dataset and frequency : the standard MinVar portfolio, the ℓ_2 portfolio with dynamic γ^* from Question 3, the ℓ_1 portfolio, and the ℓ_1 - ℓ_∞ portfolio.

TABLE 9 – Regularized strategies – 25 portfolios (5×5)

		Monthly				Daily			
Strategy		Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
25 Ptf	MinVar	17,91%	13,55%	1,322	2,016	19,87%	13,16%	1,510	0,661
	ℓ_2 (γ^* K-fold)	16,68%	12,71%	1,313	1,170	19,65%	13,01%	1,511	0,642
	ℓ_1	16,30%	13,09%	1,245	0,922	13,79%	17,80%	0,775	0,032
	ℓ_1 - ℓ_∞	15,88%	13,28%	1,196	0,561	13,62%	17,72%	0,769	0,133

TABLE 10 – Regularized strategies – 100 portfolios (10×10)

	Strategy	Monthly				Daily			
		Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
100 Ptf	MinVar	18,14%	25,57%	0,709	16,724	19,51%	11,06%	1,765	0,980
	ℓ_2 (γ^* K-fold)	17,38%	13,05%	1,332	1,671	18,95%	11,02%	1,720	0,950
	ℓ_1	17,88%	12,84%	1,392	1,426	13,81%	16,66%	0,829	0,176
	ℓ_1 - ℓ_∞	16,72%	13,09%	1,277	0,785	13,79%	13,96%	0,988	0,500

TABLE 11 – Regularized strategies – 10 industries

	Strategy	Monthly				Daily			
		Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
10 Ind.	MinVar	12,17%	12,40%	0,982	0,394	12,24%	13,40%	0,913	0,201
	ℓ_2 (γ^* K-fold)	12,27%	12,22%	1,005	0,332	12,43%	13,35%	0,931	0,215
	ℓ_1	12,33%	12,38%	0,996	0,335	12,65%	15,37%	0,823	0,320
	ℓ_1 - ℓ_∞	12,49%	12,14%	1,029	0,310	13,32%	16,17%	0,824	0,105

TABLE 12 – Regularized strategies – 48 industries

	Strategy	Monthly				Daily			
		Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
48 Ind.	MinVar	9,97%	14,23%	0,701	1,969	11,11%	12,41%	0,895	0,463
	ℓ_2 (γ^* K-fold)	11,81%	11,78%	1,003	0,685	11,55%	12,31%	0,939	0,452
	ℓ_1	11,33%	12,02%	0,942	0,775	12,90%	15,03%	0,859	0,109
	ℓ_1 - ℓ_∞	11,38%	12,00%	0,948	0,767	11,91%	14,13%	0,843	0,539

4.4 Discussion

The results show that the alternative constraints behave differently depending on the frequency of the data. With monthly returns, the ℓ_1 portfolio is competitive, especially in the 100-portfolio dataset. Its Sharpe ratio reaches 1.392, compared with 1.332 for the ℓ_2 strategy and only 0.709 for the unregularized MinVar portfolio. This suggests that controlling total exposure and allowing sparsity can be useful when the covariance matrix is noisy.

The ℓ_1 - ℓ_∞ portfolio is generally more conservative. It often delivers a lower turnover than the ℓ_1 portfolio, because the ℓ_∞ term limits the largest absolute weights. For example, for the 100 monthly portfolios, turnover falls from 1.426 for ℓ_1 to 0.785 for ℓ_1 - ℓ_∞ . The cost of this additional stability is usually a slightly lower Sharpe ratio.

The picture is different for daily data. The ℓ_1 and ℓ_1 - ℓ_∞ strategies underperform the ℓ_2 strategy in most cases. For instance, for the 25 daily portfolios, the ℓ_2 portfolio has a Sharpe ratio of 1.511, while the ℓ_1 and ℓ_1 - ℓ_∞ portfolios are around 0.77. Since daily covariance estimates are more stable, the stronger sparsity and concentration controls appear too restrictive on the tested grids.

Overall, the ℓ_2 strategy remains the most robust across datasets and frequencies. The ℓ_1 penalty can improve performance in high-dimensional monthly settings, especially for the 100 portfolios, but it is more sensitive to the calibration grid. The ℓ_1 - ℓ_∞ constraint is useful when turnover and concentration are important, but it may introduce too much bias when the covariance matrix is already well estimated.

5 Dimensionality Reduction with PCA and Stein Eigenvalue Shrinkage

5.1 PCA-Based Covariance Estimation

The objective of this section is to study whether dimensionality reduction can improve the out-of-sample performance of the mean-variance and minimum-variance portfolios. Instead of using the full sample covariance matrix $\hat{\Sigma}$, we estimate the covariance matrix using only the first K principal components.

Let

$$\hat{\Sigma} = \hat{\mathbf{V}}\hat{\Lambda}\hat{\mathbf{V}}^\top$$

be the eigendecomposition of the sample covariance matrix, where the eigenvalues are ordered as

$$\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_N.$$

Keeping only the first K principal components gives the following PCA-based covariance approximation :

$$\hat{\Sigma}_K = \hat{\mathbf{V}}_K \hat{\Lambda}_K \hat{\mathbf{V}}_K^\top + \hat{\sigma}_{\varepsilon,K}^2 \left(I_N - \hat{\mathbf{V}}_K \hat{\mathbf{V}}_K^\top \right),$$

where

$$\hat{\mathbf{V}}_K = (\hat{\mathbf{v}}_1, \dots, \hat{\mathbf{v}}_K), \quad \hat{\Lambda}_K = \text{diag}(\hat{\lambda}_1, \dots, \hat{\lambda}_K),$$

and

$$\hat{\sigma}_{\varepsilon,K}^2 = \frac{1}{N-K} \sum_{j=K+1}^N \hat{\lambda}_j$$

is the average residual variance of the discarded components. When $K = N$, no residual component is added and $\hat{\Sigma}_K = \hat{\Sigma}$.

The PCA approximation keeps the dominant systematic sources of variation while replacing the remaining directions by a common noise level. This reduces the influence of small and unstable sample eigenvalues, which are often responsible for extreme portfolio weights.

The MinVar and MSR portfolios are then computed as

$$\mathbf{w}_{\text{MinVar},K} = \frac{\hat{\Sigma}_K^{-1} \mathbf{1}}{\mathbf{1}^\top \hat{\Sigma}_K^{-1} \mathbf{1}}, \quad \mathbf{w}_{\text{MSR},K} = \frac{\hat{\Sigma}_K^{-1} \hat{\boldsymbol{\mu}}}{\mathbf{1}^\top \hat{\Sigma}_K^{-1} \hat{\boldsymbol{\mu}}}.$$

In the tables below, we focus on the MinVar results because this strategy is directly driven by the covariance matrix. The MSR portfolio is affected by the same covariance approximation, but it also remains highly sensitive to the estimation error in the mean vector $\hat{\boldsymbol{\mu}}$.

5.2 Choosing the Number of Principal Components

The number of retained components K controls the bias–variance trade-off. A small K gives a more stable but more restrictive factor representation, while a large K is closer to the full covariance matrix but may reintroduce estimation noise.

Instead of relying only on a visual scree plot or on an arbitrary percentage of explained variance, we use the Bai and Ng information criterion :

$$IC_2(K) = \log \left(\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \hat{\varepsilon}_{it,K}^2 \right) + K \frac{N+T}{NT} \log \left(\frac{NT}{N+T} \right),$$

where

$$\hat{\varepsilon}_K = X - \hat{\mathbf{V}}_K \hat{\mathbf{V}}_K^\top X$$

is the reconstruction error after keeping the first K principal components. The selected value K^* is the value that minimizes this criterion over the candidate grid.

The following adaptive grids are used :

- $N = 10$: $\mathcal{K} = \{1, 2, 3, 5, 8, 10\}$;
- $N = 25$: $\mathcal{K} = \{1, 2, 3, 5, 8, 10, 15, 20, 24, 25\}$;
- $N = 48$: $\mathcal{K} = \{1, 2, 3, 5, 8, 10, 15, 20, 24, 30, 40, 48\}$;
- $N = 100$ monthly : $\mathcal{K} = \{1, 2, 3, 5, 8, 10, 15, 20, 25, 30\}$;
- $N = 100$ daily : $\mathcal{K} = \{1, 2, 3, 5, 8, 10, 15, 20, 30, 50, 75, 100\}$.

5.3 Stein Shrinkage of Eigenvalues

We also compare the sample eigenvalues with Stein-corrected eigenvalues. The motivation is that the eigenvalues of the sample covariance matrix can be distorted when the number of assets is large relative to the sample size : the largest sample eigenvalues tend to be overestimated, while the smallest ones tend to be underestimated.

Stein’s approach keeps the sample eigenvectors but replaces the sample eigenvalues by corrected values :

$$\tilde{\Sigma} = \hat{\mathbf{V}} \tilde{\Lambda} \hat{\mathbf{V}}^\top.$$

In the classical Stein estimator, the corrected eigenvalues are obtained from

$$\tilde{\lambda}_i = \frac{T\hat{\lambda}_i}{\alpha_i}, \quad \alpha_i = T - N + 1 + 2\hat{\lambda}_i \sum_{j \neq i} \frac{1}{\hat{\lambda}_i - \hat{\lambda}_j}.$$

In practice, the corrected eigenvalues may need to be adjusted to preserve positivity and ordering. The purpose is to obtain a better-conditioned covariance matrix and reduce the instability of portfolio weights. We therefore compare the out-of-sample performance obtained with sample eigenvalues and with Stein-corrected eigenvalues.

5.4 Results

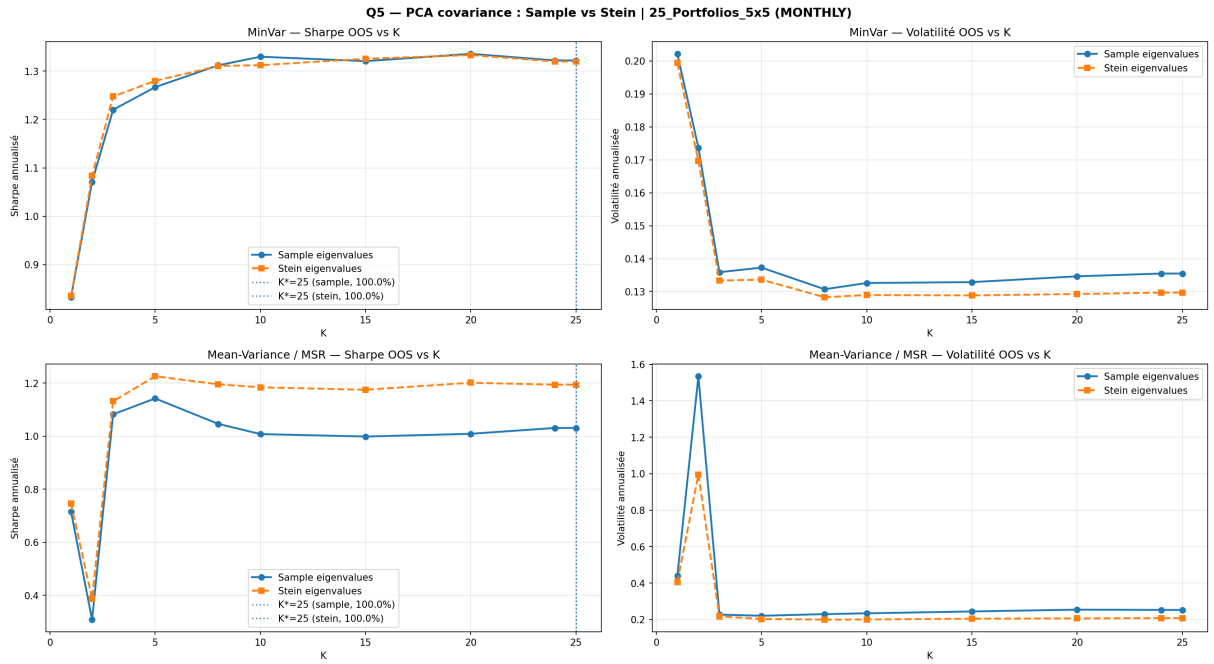


FIGURE 2 — Out-of-sample Sharpe ratio as a function of K – Sample vs Stein eigenvalues, 25 portfolios, monthly data. The vertical dashed line indicates the K^* selected by Bai and Ng.

The following tables report the out-of-sample Sharpe ratio of the PCA-MinVar portfolio as a function of K , using either sample eigenvalues or Stein-corrected eigenvalues. Turnover is also reported for the sample-eigenvalue version.

TABLE 13 – PCA-MinVar : out-of-sample Sharpe ratio as a function of K – 25 portfolios

K	Monthly			Daily		
	SR Sample	SR Stein	TO Sample	SR Sample	SR Stein	TO Sample
1	0,833	0,835	0,336	1,050	1,050	0,364
2	1,071	1,083	0,704	1,229	1,229	0,356
3	1,220	1,248	0,865	1,338	1,338	0,499
5	1,267	1,280	1,167	1,394	1,394	0,674
8	1,312	1,311	1,484	1,408	1,410	0,653
10	1,330	1,312	1,703	1,450	1,452	0,666
15	1,321	1,325	1,942	1,495	1,495	0,690
20	1,336	1,333	2,028	1,518	1,513	0,670
25 (K^*)	1,322	1,320	2,016	1,510	1,506	0,661

TABLE 14 – PCA-MinVar : out-of-sample Sharpe ratio as a function of K – 100 portfolios

K	Monthly			Daily		
	SR Sample	SR Stein	TO Sample	SR Sample	SR Stein	TO Sample
1	0,845	0,878	0,430	1,098	1,097	0,394
3	1,228	1,253	1,010	1,374	1,372	0,578
5 (K^* Sample)	1,306	1,316	1,317	1,380	1,382	0,685
8	1,316	1,314	1,600	1,545	1,544	0,855
10	1,394	1,363	1,692	1,632	1,629	0,859
15	1,375	1,350	2,035	1,650	1,647	0,887
20	1,359	1,353	2,279	1,630	1,632	0,942
3 (K^* Stein)	1,228	1,253	1,010	–	–	–
100 (K^* Daily)	–	–	–	1,765	1,744	0,980

TABLE 15 – PCA-MinVar : out-of-sample Sharpe ratio as a function of K – 10 and 48 industries

Dataset	K	SR Sample	SR Stein	Vol. Sample	TO Sample
10 Ind. (M)	3	0,928	0,935	0,126	0,323
	5	0,954	0,963	0,125	0,361
	8	0,982	0,983	0,125	0,395
	10 (K^*)	0,982	0,982	0,124	0,394
48 Ind. (M)	3	1,043	1,054	0,124	0,474
	8	0,990	1,044	0,125	0,742
	10	0,982	1,038	0,127	0,797
	20	0,841	0,995	0,130	1,243
	48 (K^*)	0,701	0,943	0,142	1,969
48 Ind. (D)	8	0,883	0,886	0,128	0,412
	15	0,912	0,916	0,127	0,435
	48 (K^*)	0,895	0,901	0,124	0,463

TABLE 16 – K^* selected by Bai and Ng and corresponding MinVar performance

Dataset	Freq.	K^* Sample	SR Sample	K^* Stein	SR Stein
25 Ptf	M	25	1,322	25	1,320
25 Ptf	D	25	1,510	25	1,506
100 Ptf	M	5	1,306	3	1,253
100 Ptf	D	100	1,765	100	1,744
10 Ind.	M	10	0,982	10	0,982
10 Ind.	D	10	0,913	10	0,913
48 Ind.	M	48	0,701	48	0,943
48 Ind.	D	48	0,895	48	0,901

M = monthly ; D = daily ; SR = out-of-sample Sharpe ratio.

Summary of selected K^* values.

5.5 Discussion

The results show that PCA-based covariance estimation is useful mainly when the covariance matrix is noisy and high-dimensional. For the 100 monthly portfolios, the unregularized MinVar portfolio from Question 1 was unstable, with a Sharpe ratio of 0.709 and a very high turnover. By keeping a small number of principal components, the Sharpe ratio increases substantially. With $K = 5$, the Sharpe ratio reaches 1.306, and with $K = 10$ it reaches 1.394. This confirms that removing the smallest and noisiest eigenvalues can strongly improve out-of-sample robustness.

For the 25 portfolios and the 10 industries, the effect of PCA is more limited. The Bai and Ng criterion often selects $K^* = N$, meaning that the full covariance matrix is retained.

In these cases, the sample covariance matrix is already sufficiently stable, so dimension reduction does not bring a large improvement. This is also visible in the fact that the Sharpe ratios for large K are close to the benchmark MinVar results.

The results for the 48 monthly industries are more nuanced. Bai and Ng selects $K^* = 48$, so PCA alone does not reduce the dimension. However, Stein shrinkage improves the Sharpe ratio from 0.701 to 0.943. This suggests that the main issue is not necessarily the number of retained components, but the distortion of the sample eigenvalues. Correcting the eigenvalues improves the conditioning of the covariance matrix and leads to more stable portfolio weights.

Daily data require less dimension reduction. Since the number of observations is much larger, the covariance matrix is estimated more accurately. This explains why Bai and Ng usually selects the full set of components for daily returns, especially for the 100 portfolios. In this case, using all components with sample eigenvalues gives a Sharpe ratio of 1.765, while Stein gives a slightly lower value of 1.744.

Overall, PCA helps when the dataset is large relative to the estimation window, as in the 100 monthly portfolios. Stein's eigenvalue correction provides additional robustness, but its gains are dataset-dependent. It is particularly useful when the full set of components is retained but the empirical eigenvalues remain distorted, as observed for the 48 monthly industries. For the MSR portfolio, the same covariance approximation can reduce sensitivity to small eigenvalues, but it does not eliminate the main source of instability, namely the estimation error in expected returns.

6 IC-Variance-Parity with FastICA

6.1 Methodology

The FastICA algorithm decomposes the return matrix into **independent components** (ICs). Unlike PCA, which produces uncorrelated components, ICA aims to extract components that are statistically independent, especially in terms of higher-order moments. This property is particularly relevant in finance, where asset returns often exhibit fat tails and skewness.

The IC-Variance-Parity portfolio assigns an equal contribution to portfolio variance across the independent components :

$$\mathbf{w}_{\text{ICVP}} = \mathbf{A}\mathbf{w}_{\text{VP}}, \quad (\mathbf{w}_{\text{VP}})_k = \frac{1/\sigma_k^2}{\sum_{j=1}^K 1/\sigma_j^2},$$

where $\mathbf{A}$ is the mixing matrix estimated by FastICA and σ_k^2 is the variance of the k -th independent component.

6.2 Evolution of Moments as a Function of K

The tables below compare, for each dataset and frequency, the six risk metrics of MinVar, EW and the selected ICVP portfolios.

TABLE 17 – ICVP vs MinVar vs EW – 25 portfolios (5×5)

Freq.	Strategy	Mean	Vol.	SR	Skew	ExKurt	MPaR
M	MinVar	17,91%	13,55%	1,322	−0,264	0,827	0,312
	EW	13,55%	17,62%	0,769	−0,753	2,709	0,542
	ICVP $K = 1$	13,38%	18,23%	0,734	−0,715	2,564	0,555
	ICVP $K = 5$	15,70%	19,16%	0,819	−0,321	1,623	0,511
	ICVP $K = 8$	15,61%	16,61%	0,940	−0,288	2,540	0,470
	ICVP $K = 10$	16,10%	16,41%	0,982	−0,253	1,310	0,412
	ICVP $K = 15$	16,52%	16,16%	1,022	−0,320	1,304	0,409
D	MinVar	19,87%	13,16%	1,510	−0,288	24,285	1,064
	EW	13,59%	18,30%	0,743	−0,592	12,206	0,995
	ICVP $K = 8$	16,97%	17,86%	0,950	+0,181	9,218	0,764
	ICVP $K = 15$	18,44%	17,58%	1,049	+0,171	7,233	0,671
	ICVP $K = 20$	19,79%	18,81%	1,052	+2,216	114,76	4,817

TABLE 18 – ICVP vs MinVar vs EW – 100 portfolios (10×10)

Freq.	Strategy	Mean	Vol.	SR	Skew	ExKurt	MPaR
M	MinVar	18,14%	25,57%	0,709	−0,140	0,096	0,573
	EW	13,63%	17,80%	0,766	−0,743	2,762	0,550
	ICVP $K = 5$	15,79%	18,14%	0,871	−0,384	2,017	0,503
	ICVP $K = 10$	13,58%	16,82%	0,808	−0,541	2,233	0,488
D	MinVar	19,51%	11,06%	1,765	−1,006	16,278	0,705
	EW	13,67%	18,31%	0,747	−0,600	12,121	0,992
	ICVP $K = 20$	16,62%	16,69%	0,995	−0,170	6,918	0,667

TABLE 19 – ICVP vs MinVar vs EW – 10 and 48 industries

Dataset	Strategy	Mean	Vol.	SR	Skew	ExKurt	MVaR
10 Ind. (M)	MinVar	12,17%	12,40%	0,982	−0,319	0,494	0,292
	EW	13,12%	14,70%	0,893	−0,579	2,448	0,432
	ICVP $K = 3$	12,81%	15,66%	0,818	−0,432	1,261	0,412
	ICVP $K = 8$	10,85%	14,87%	0,729	−0,156	0,207	0,338
48 Ind. (M)	MinVar	9,97%	14,23%	0,701	−0,161	0,893	0,348
	EW	13,03%	16,40%	0,795	−0,686	3,236	0,522
	ICVP $K = 10$	14,60%	17,13%	0,852	−0,143	1,177	0,420
	ICVP $K = 15$	15,87%	16,06%	0,989	−0,077	1,530	0,394

SR = Sharpe ratio with $r_f = 0$; ExKurt = excess kurtosis; MVaR = Cornish–Fisher modified VaR at the 1% level.

6.3 Cornish–Fisher Modified Value-at-Risk

The MVaR accounts for skewness S and excess kurtosis κ through the Cornish–Fisher expansion :

$$\text{MVaR}_{1\%} = -(\mu_{\text{ann}} + \sigma_{\text{ann}} z_{\text{CF}}), \quad z_{\text{CF}} = z_{\alpha} + \frac{z_{\alpha}^2 - 1}{6}S + \frac{z_{\alpha}^3 - 3z_{\alpha}}{24}\kappa - \frac{2z_{\alpha}^3 - 5z_{\alpha}}{36}S^2$$

where $z_{\alpha} = \Phi^{-1}(0,01) \approx -2,326$.

6.4 Discussion

The ICVP strategy mainly improves the higher-moment profile of the portfolio, especially for daily data. For the 25 daily portfolios, excess kurtosis decreases sharply from 24.285 for MinVar to 7.233 for ICVP with $K = 15$, while MVaR falls from 1.064 to 0.671. A similar pattern appears for the 100 daily portfolios, where excess kurtosis decreases from 16.278 to 6.918 and skewness improves from -1.006 to -0.170 .

These improvements generally come with a lower Sharpe ratio than MinVar. For example, on the 25 monthly portfolios, the best ICVP specification has a Sharpe ratio of 1.022, compared with 1.322 for MinVar. This is expected : ICVP is not designed to minimize variance directly, but to diversify risk across independent components and improve tail-risk properties.

The results are more favourable for some monthly datasets. On the 100 monthly portfolios, ICVP with $K = 5$ improves the Sharpe ratio relative to MinVar, from 0.709 to 0.871, and reduces MVaR from 0.573 to 0.503. For the 48 monthly industries, ICVP with $K = 15$ also improves the Sharpe ratio, from 0.701 to 0.989, and makes skewness less negative. However,

in that case, excess kurtosis and MVaR are higher than for MinVar, so the improvement is not uniform across all risk measures.

The choice of K is therefore crucial. A value that is too small may ignore relevant risk sources, whereas a value that is too large may introduce noisy independent components. This is visible for the 25 daily portfolios : ICVP with $K = 20$ achieves a similar Sharpe ratio to $K = 15$, but produces an extreme excess kurtosis of 114.76 and a very high MVaR of 4.817. Hence, K should not be selected only by maximizing the Sharpe ratio ; skewness, excess kurtosis and MVaR must also be monitored.

TABLE 20 – Relative change of ICVP (K^*) compared with MinVar

Dataset	Freq.	K^* ICVP	Δ SR	Δ Skew	Δ ExKurt	Δ MVaR
25 Ptf	M	15	-0,300	-0,056	+0,477	+0,097
25 Ptf	D	15	-0,461	+0,459	-17,052	-0,393
100 Ptf	M	5	+0,162	-0,244	+1,921	-0,070
100 Ptf	D	20	-0,770	+0,836	-9,360	-0,038
10 Ind.	M	8	-0,253	+0,163	-0,287	+0,046
48 Ind.	M	15	+0,288	+0,084	+0,637	+0,046

Differences are computed as ICVP minus MinVar. For SR and skewness, a positive value is favourable.

For excess kurtosis and MVaR, a negative value is favourable.

Comparison of ICVP and MinVar on key moments.

7 Additional Strategy : OSCAR Minimum-Variance Portfolio

7.1 Presentation and Justification

The additional strategy considered is the **OSCAR minimum-variance portfolio** (*Octagonal Shrinkage and Clustering Algorithm for Regression*). The portfolio minimizes variance under an OSCAR-type norm constraint :

$$\mathbf{w}^* = \underset{\mathbf{1}^\top \mathbf{w} = 1}{\operatorname{argmin}} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \quad \text{s.t.} \quad \|\mathbf{w}\|_1 + \alpha \sum_{k=1}^{N-1} \|\mathbf{w}\|_{(k)} \leq c,$$

where $\|\mathbf{w}\|_{(k)}$ denotes the sum of the k largest absolute portfolio weights.

Objective : weight clustering. The main property of the OSCAR norm is that it encourages assets with similar risk profiles to receive similar or identical weights. This clustering effect is economically meaningful, especially for industry portfolios. Closely related sectors are likely to share common risk factors, so grouping them through similar weights can reduce unstable long-short positions and improve robustness.

Parameters. The parameters are fixed at $\alpha = 0.02$ and $c = 1.5 \times c_{\min}$, where

$$c_{\min} = 1 + \alpha \frac{N - 1}{2}$$

is the OSCAR norm value of the equally weighted portfolio. The effective values of c are :

— $N = 10 : c = 1.635$; $N = 25 : c = 1.860$; $N = 48 : c = 2.205$; $N = 100 : c = 2.985$.

This additional strategy is based on the methodology presented in Section 2.2 of Xing, Hu and Yang , *Robust minimum variance portfolio with L-infinity constraints*. This paper is one of the reference articles made available on Moodle for the course, which makes it directly relevant for the additional strategy required in this question.

7.2 Out-of-Sample Results

TABLE 21 – OSCAR vs MinVar vs EW – monthly and daily data

Dataset	Strategy	Monthly				Daily			
		Mean	Vol.	SR	TO	Mean	Vol.	SR	TO
25 Ptf	EW	13,55%	17,62%	0,769	0,000	13,59%	18,30%	0,743	0,000
	MinVar	17,91%	13,55%	1,322	2,016	19,87%	13,16%	1,510	0,661
	OSCAR	14,50%	14,03%	1,034	0,289	16,10%	15,08%	1,068	0,195
100 Ptf	EW	13,63%	17,80%	0,766	0,000	13,67%	18,31%	0,747	0,000
	MinVar	18,14%	25,57%	0,709	16,724	19,51%	11,06%	1,765	0,980
	OSCAR	13,94%	14,73%	0,947	0,325	15,09%	14,76%	1,022	0,223
10 Ind.	EW	13,12%	14,70%	0,893	0,000	13,33%	16,62%	0,802	0,000
	MinVar	12,17%	12,40%	0,982	0,394	12,24%	13,40%	0,913	0,201
	OSCAR	12,07%	12,15%	0,993	0,232	11,74%	13,32%	0,882	0,136
48 Ind.	EW	13,03%	16,40%	0,795	0,000	13,19%	17,22%	0,766	0,000
	MinVar	9,97%	14,23%	0,701	1,969	11,11%	12,41%	0,895	0,463
	OSCAR	11,55%	12,18%	0,949	0,278	11,53%	12,97%	0,889	0,150

7.3 Observed Clustering

TABLE 22 – Average number of clusters formed by OSCAR across rolling windows

Dataset	N	α	Average clusters
25 Ptf (M and D)	25	0,020	5,52 (M) / 6,04 (D)
100 Ptf (M and D)	100	0,020	7,63 (M) / 8,59 (D)
10 Ind. (M and D)	10	0,020	7,62 (M) / 7,82 (D)
48 Ind. (M and D)	48	0,020	11,17 (M) / 12,34 (D)

7.4 Discussion

The main advantage of OSCAR is the strong reduction in turnover. Across all datasets and frequencies, OSCAR produces turnovers between roughly 0.14 and 0.33, compared with much higher values for the unconstrained MinVar portfolio. The most striking case is the 100 monthly portfolios, where turnover falls from 16.724 for MinVar to only 0.325 for OSCAR. This confirms that the clustering constraint strongly stabilizes the portfolio weights.

In terms of Sharpe ratio, OSCAR generally lies between EW and MinVar. For the 25 and 100 monthly portfolios, OSCAR clearly improves over EW, with Sharpe ratios of 1.034 and 0.947, compared with 0.769 and 0.766 for EW. However, it remains below the best regularized MinVar specifications obtained in Questions 3 and 4. This is expected, since OSCAR introduces an additional clustering constraint and therefore accepts some bias in exchange for lower turnover.

The strategy performs particularly well for industry portfolios. For the 10 industries, OSCAR obtains a monthly Sharpe ratio of 0.993, close to MinVar’s 0.982, while reducing turnover from 0.394 to 0.232. For the 48 industries, OSCAR improves the monthly Sharpe ratio from 0.701 for MinVar to 0.949, while reducing turnover from 1.969 to 0.278. This suggests that the clustering mechanism is useful when assets are naturally grouped into related sectors.

The average number of clusters is much smaller than the number of assets. For instance, the 48 industry portfolios are reduced to around 11–12 clusters on average, while the 100 size-book-to-market portfolios are reduced to around 8 clusters. This indicates that OSCAR captures a lower-dimensional structure in the asset universe. Economically, these clusters can be interpreted as groups of assets with similar exposure to common risk factors, although the exact interpretation would require inspecting the cluster composition.

Overall, OSCAR is not always the best strategy in terms of raw Sharpe ratio, but it offers a very attractive compromise between performance and implementability. Its main strength is the reduction of turnover and weight instability. Once transaction costs are considered, this lower turnover could make OSCAR competitive with strategies that achieve higher gross Sharpe ratios but require much more frequent and aggressive rebalancing.

8 Three Main Conclusions

The analysis leads to three main conclusions.

1. **In-sample performance is a poor indicator of true portfolio quality.** The MSR portfolio often performs best in-sample, but its out-of-sample performance is unstable because it relies heavily on noisy estimates of expected returns. This instability becomes particularly severe in high-dimensional datasets, where the turnover can become extremely large. Simpler strategies such as EW and MinVar

are therefore often more robust out-of-sample.

2. **Regularization improves the Sharpe–turnover trade-off.** Penalizing portfolio weights, especially through the ℓ_2 constraint, stabilizes the covariance inversion and reduces extreme allocations. The improvement is particularly clear for the 100 monthly portfolios, where the Sharpe ratio increases substantially while turnover falls sharply. Stronger constraints such as OSCAR further reduce turnover, although sometimes at the cost of a lower gross Sharpe ratio.
3. **Dimension, frequency and higher moments matter.** Daily data generally improve covariance estimation, while PCA and Stein shrinkage can help when eigenvalues are noisy. In addition, ICVP shows that looking beyond variance is important : skewness, excess kurtosis and modified VaR provide useful information about tail risk that is not captured by the Sharpe ratio alone.

9 Best Practices for Out-of-Sample Performance

Based on the empirical results, the following best practices should be used to obtain robust out-of-sample portfolio performance :

1. **Avoid unregularized MSR portfolios**, because expected returns are very noisy and lead to unstable weights.
2. **Regularize the covariance matrix** when the number of assets is large relative to the estimation window.
3. **Select hyperparameters by cross-validation**, using only the estimation window and never the out-of-sample period.
4. **Use daily data to estimate covariance matrices** when available, while keeping a reasonable rebalancing frequency.
5. **Control turnover and extreme positions** through weight constraints such as ℓ_2 , ℓ_1 or OSCAR.
6. **Evaluate portfolios with several risk measures**, not only the Sharpe ratio, but also turnover, skewness, excess kurtosis and modified VaR.

10 Reformulation of the Standard Warning

The standard warning can be made more accurate by explicitly mentioning estimation error and the distinction between in-sample and out-of-sample performance :

“Past performance is no guarantee of future results ; only robust out-of-sample performance, obtained with controlled estimation error and realistic rebalancing assumptions, provides a more credible but still imperfect indication of a strategy’s future performance.”

10 Bonus Question : Optimized Strategy on 300 Stocks (2010–2019)

10.0 Data and Objective

The available data consist of :

- **Daily returns** : 7 560 observations $\times$ 300 stocks (Jan. 1980 – Dec. 2009) ;
- **Monthly returns** : 348 observations $\times$ 300 stocks.

The objective is to submit a weight vector $\mathbf{w}^* \in \mathbb{R}^{300}$ computed from these data, with the aim of maximizing the out-of-sample Sharpe ratio over the hidden period **January 2010 – December 2019**.

10.0 Selected Strategy : Long-Only Regularized Mean-Variance Portfolio

The selected strategy is a long-only regularized mean-variance portfolio :

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \hat{\boldsymbol{\mu}}_s^\top \mathbf{w} - \rho \mathbf{w}^\top \hat{\boldsymbol{\Sigma}} \mathbf{w} - \eta \|\mathbf{w}\|_2^2, \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}, \quad w_i \leq w_{\max},$$

where $\hat{\boldsymbol{\mu}}_s = \delta_s \hat{\boldsymbol{\mu}}$ denotes the shrunk expected-return vector. A total of 144 parameter combinations were compared using an internal rolling-window backtest on 1980–2009, with a 10-year estimation window and a 1-year rebalancing step.

TABLE 23 – Top 10 parameter combinations – internal grid search based on OOS Sharpe ratio

δ_s	ρ	η	$w_{\max}$	Mean	Vol.	SR	SR _{EW}
0,30	5	0,0100	3%	19,55%	15,89%	1,230	1,241
0,30	5	0,0010	3%	19,58%	15,94%	1,228	1,241
0,30	5	0,0001	3%	19,58%	15,95%	1,228	1,241
0,30	5	0,0100	5%	19,53%	16,56%	1,179	1,241
0,50	5	0,0100	3%	18,83%	16,04%	1,174	1,241
0,30	5	0,0010	5%	19,57%	16,68%	1,173	1,241
0,30	5	0,0001	5%	19,57%	16,70%	1,172	1,241
0,50	5	0,0010	3%	18,80%	16,15%	1,165	1,241
0,50	5	0,0001	3%	18,80%	16,16%	1,164	1,241
0,50	5	0,0100	5%	19,23%	16,68%	1,153	1,241

TABLE 24 – Selected parameters and internal OOS performance

Parameter	Tested Grid	Selected Value
δ_s (mean shrinkage)	{0,1; 0,3; 0,5; 0,7; 0,9}	0,30
ρ (risk aversion)	{1; 5; 10; 20; 40}	5,0
η (Ridge penalty)	$\{10^{-4}; 10^{-3}; 10^{-2}; 10^{-1}\}$	0,01
$w_{\max}$ (maximum weight)	{1%; 3%; 5%; 8%; 10%}	3%

10.0 Internal OOS Performance and Portfolio Characteristics

TABLE 25 – OOS performance from the 1980–2009 backtest

Strategy	Ann. Mean	Ann. Vol.	Sharpe
Regularized MV ($\mathbf{w}^*$)	19,55%	15,89%	1,230
Equally weighted (EW)	18,01%	14,51%	1,241

TABLE 26 – Characteristics of the submitted weight vector

Sum of weights	1,000
Minimum weight	0,000 (long-only)
Maximum weight	3,000%
Active assets ($w_i > 0$)	84 out of 300

10.0 Comment

A notable result is that the equally weighted portfolio has a slightly higher internal Sharpe ratio than the selected regularized MV strategy (1,241 versus 1,230). This again illustrates one of the central findings of the project : simple EW portfolios are difficult to beat consistently, even when using a carefully regularized optimization procedure.

Nevertheless, the selected strategy differs from EW in two important ways. First, it selects only **84 assets out of 300**, producing a sparse and more selective portfolio. Second, it incorporates a mild directional signal through the shrunk expected-return vector, with $\delta_s = 0,3$. Although this signal is not strong enough to outperform EW in the internal 1980–2009 backtest, it may add value over the hidden 2010–2019 period if return dispersion across stocks and sectors is sufficiently high.

The 300 submitted weights $\mathbf{w}^*$ are provided in `portfolio_weights_bonus.xlsx`.

A Implementation Details

The full code is implemented in **Python 3.10** using Google Colab. The main libraries used are :

- `pandas` and `numpy` : data manipulation and matrix computations.
- `pandas_datareader` : download of Fama–French datasets.
- `scipy.stats` : distribution functions, especially for the Cornish–Fisher modified VaR.
- `cvxpy` : convex optimization for portfolios with ℓ_1 and ℓ_1 - ℓ_∞ constraints.
- `sklearn.decomposition.FastICA` : FastICA algorithm for Question 6.
- `sklearn.model_selection.KFold` : cross-validation for hyperparameter selection.
- `matplotlib` : visualization of the results.

A.0 Rolling-Window Parameters

TABLE 27 – Rolling-window parameters

Frequency	Estimation Window	Step
Monthly	120 months (10 years)	6 months
Daily	2520 days (10 years)	126 days

B Data

TABLE 28 – Fama–French datasets

Dataset	N	Period	Frequencies
25 Portfolios (Size $\times$ B/M)	25	Jan. 1970 – Dec. 2025	M, D
100 Portfolios (Size $\times$ B/M)	100	Jan. 1970 – Dec. 2025	M, D
10 Industry Portfolios	10	Jan. 1970 – Dec. 2025	M, D
48 Industry Portfolios	48	Jan. 1970 – Dec. 2025	M, D

M = monthly, D = daily. Missing values coded as -99.99 or -999 are replaced by 0.